Kavin Nandha M K

Software Engineer

+91 93455 69707 | kavinnandhakavin@gmail.com | linkedin.com/in/kavinnandha | github.com/kavinnandha
kavin.codegarage.dev | Coimbatore, India | Open to relocation | Available for a 6-month internship (Jan–Jun 2027)

SUMMARY

Final-year CSE (Cyber Security) student who shipped Node.js features, MongoDB query optimizations, and JWT/RBAC authentication to production during a six-month DevOps & SDE internship. Deployed a bare-metal Kubernetes (k3s) platform that sustained 2,500+ concurrent users. Strong in data structures, algorithms, OOP, and DBMS.

EDUCATION

B.E. Computer Science and Engineering (Cyber Security) | CGPA: **8.67**/10 **2023 – 2027**
Sri Shakthi Institute of Engineering and Technology, Coimbatore
Higher Secondary Certificate (HSC), VVM Matric Hr. Sec. School, Coimbatore | 83.4% 2023

TECHNICAL SKILLS

Languages: JavaScript, TypeScript, Python, SQL, Bash
Core CS: Data Structures & Algorithms, OOP, DBMS, Operating Systems, Computer Networks
Web Development: Node.js, Express.js, REST API design, JWT/RBAC, React, Tailwind CSS
Databases: MongoDB (schema design, query optimization), MySQL, Redis
Cloud & DevOps: Docker, Kubernetes (k3s), GitHub Actions (CI/CD), AWS, Linux, Nginx, Git
AI Engineering: LLM API integration (Claude, OpenAI, Gemini), retrieval-augmented generation (RAG)

EXPERIENCE

DevOps & SDE Intern | TeamMistake Technologies **Oct 2025 – Mar 2026**
- Shipped production Node.js/Express features end to end, from data modeling and API design to deployment
- Designed MongoDB schemas and rewrote queries to remove redundant database calls on high-traffic endpoints
- Built JWT authentication and role-based access control (RBAC) from scratch for user and admin workflows
- Containerized services with Docker, shipped releases via GitHub Actions CI/CD, and managed DNS and SSL/TLS

PROJECTS

HR Connect – Placement & Recruitment Platform | *React, TypeScript, Node.js, Express.js, MongoDB, Redis, Docker*
- Built a full-stack placement platform for job postings, applications, and candidate management across 3 roles
- Designed the REST API and MongoDB data model for postings, applications, and profiles, secured with JWT/RBAC
- Added a Redis cache with invalidation on write to cut repeated database reads on frequently used endpoints
- Containerized modular backend services with Docker for reproducible development and deployment

Production Moodle LMS on Bare-Metal Kubernetes (k3s) | *k3s, Docker, MySQL, Prometheus, Grafana*
- Deployed an institution-wide Moodle LMS with CodeRunner/Jobe code grading on a bare-metal k3s cluster (dual Intel Xeon Gold 6430); sustained **2,500+ concurrent users** at peak
- Configured Deployments, Secrets, and PersistentVolumeClaims, isolating grading workloads from the web tier
- Diagnosed networking, storage, and resource-contention issues using Prometheus, Grafana, and Kubernetes logs

Sentinel AI – Cyberbullying Detection Platform | *Node.js, Express.js, MongoDB, ML integration*
- Built REST APIs and an ingestion pipeline that feeds user posts to an ML classifier to flag, escalate, and report harmful content in real time, with RBAC and audit logging
- Won SRCAS 2.0 Hackathon (350+ participants); co-authored the research paper published in JRANNS (Apr 2026)

Multilingual NCERT Doubt-Solver – Intel Unnati Industrial Training | *OPEA, RAG, LLMs* Oct – Dec 2025
- Built a multilingual question-answering assistant over NCERT textbooks using an OPEA-based RAG pipeline

Health App – Backend | *Node.js, Express.js, MongoDB*
- Built REST APIs for health tracking, reminders, and records with validation and per-user authorization

ACHIEVEMENTS & CERTIFICATIONS

- **MobilityX 2.0 Hackathon (2025):** Runner-up and Software Track Winner among 250+ participants
- **Finalist:** Samsung Solve for Tomorrow 2026 (Top 20) | MAHE Mobility Challenge 2026, AI Track (MIT Bengaluru)
- **Publication:** Co-author, *Sentinel AI: Cyber Bully Detection and Prevention*, Journal of Research in Artificial Neural Network Systems (JRANNS), Vol. 2, No. 2, Apr 2026
- **Certifications & Training:** Intel Unnati Industrial Training (2025) | AWS Cloud Practitioner Essentials, AWS Skill Builder (2026) | Claude Code in Action, Anthropic (2026) | Aspire Leaders Program (2026) | View certificates